



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KINGSWAY TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

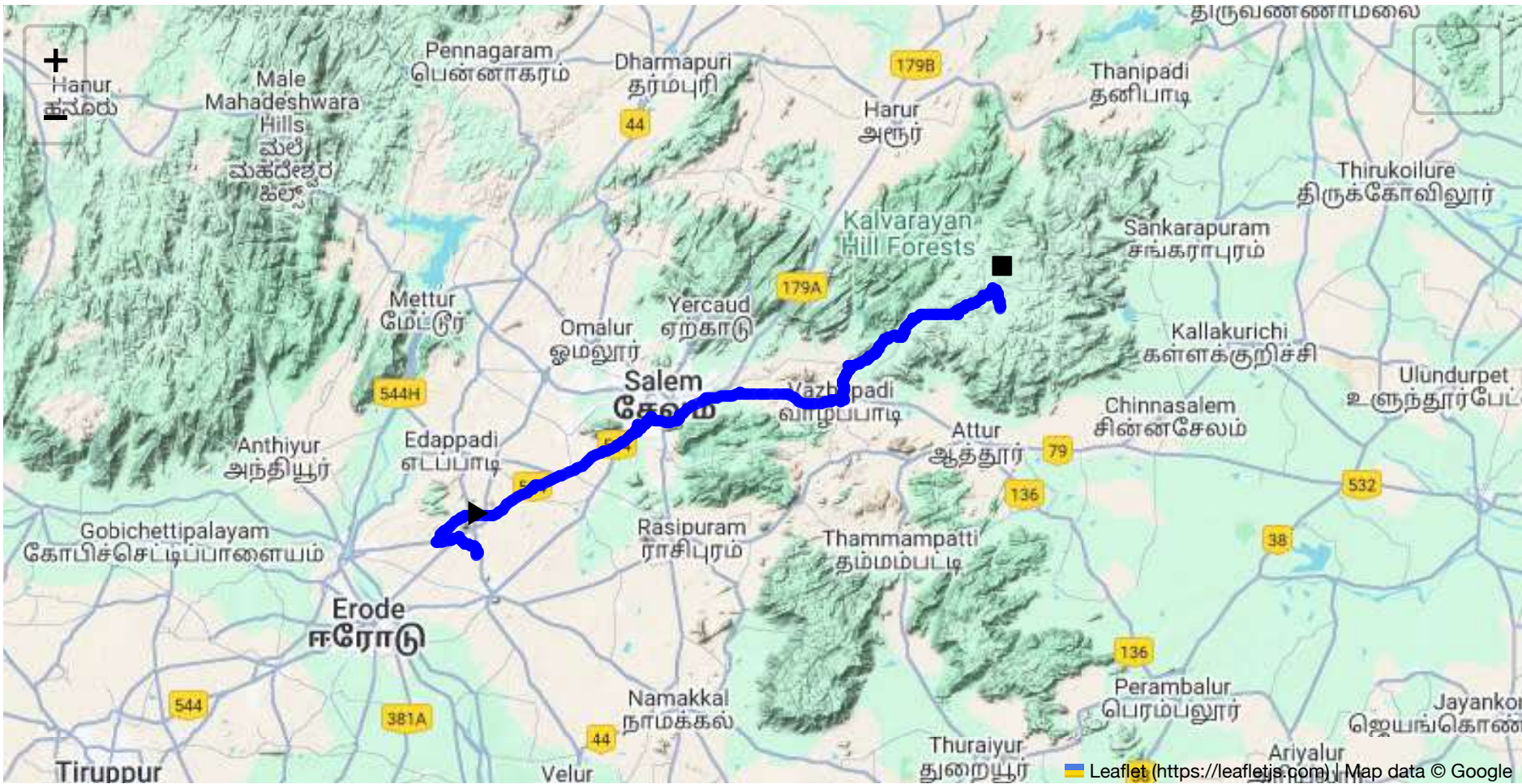
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 124.32 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours

Start: (11.4381, 77.8734)
End: (11.7877, 78.6318)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route begins at CVQF+23W, Sangagiri, and travels through several key points—namely Padaiveedu, Sandaipet, Kondalampatti, and P M P Nagar in Salem—before reaching the destination at QJQJ+7X2, Chinnakalrayan Hills Therkkunadu. The route predominantly traverses Tamil Nadu's state highways with a portion on the Bhavani and Sankari Main Roads. The entirety of the route is approximately 124.32 kilometers, aligning mostly with primary and secondary roads that facilitate transport across rural towns and suburban regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences a tropical climate with the northeast monsoon season from October to December, which can bring heavy rains and potential flooding, especially on rural roads. Temperatures can rise significantly between March and May, and this may impact road conditions. During the monsoon, reduced visibility and slippery roads are critical concerns.

3. Analysis of Traffic Patterns: Peak traffic hours are typically in the morning (7:30 AM - 10:00 AM) and evening (5:30 PM - 8:00 PM) due to local commutes. The most congested areas likely lie around Salem city, especially around Kondalampatti and Sandaipet, given their connectivity with major city hubs.

4. Assessment of Road Quality and Infrastructure: Road quality varies along the route. Urban sections, such as those in Salem, tend to be better maintained, while rural stretches may have narrower roads and less consistent maintenance. Potholes and unmarked speed bumps are common in less trafficked areas.

5. Suggestions for Alternative Routes for Emergencies: An alternative could utilize major highways like NH-44 or SH-20 bypassing heavily congested city routes. In emergencies, drivers could divert towards NH-544 or smaller feeder roads if main arteries are blocked.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transporting hazardous materials requires adherence to the guidelines set by the Petroleum and Explosives Safety Organization (PESO) in India, which mandates proper labeling, documentation, and transport during permitted hours to minimize risk and ensure public safety.

7. **Overview of Historical Incidents:** While specific historical data is limited, heavy vehicle accidents due to driver fatigue, mechanical failures, and road quality are noted concerns, with incidents generally occurring in or near urban areas during peak hours or adverse weather conditions.
8. **Environmental Considerations and Sensitive Areas:** The route passes near agricultural land and small forests, particularly around the Chinnakalrayan Hills. Sustainable driving practices are imperative to minimize environmental impact in these sensitive zones, including monitoring for wildlife crossing and runoff affecting local water sources.
9. **Analysis of Communication Coverage:** Cellular coverage is generally robust along main roads and through urbanized areas like Salem. However, in more isolated rural sections and near the Chinnakalrayan Hills, there may be occasional dead zones.
10. **Estimated Emergency Response Times:** Emergency response times vary, with urban sections near Salem potentially achieving under 30-minute response times, while rural areas may experience delays up to an hour or more, depending on accessibility and hazard severity.
12. **Overall Summary of Risk Assessment:** The primary risks along this route include weather-related hazards during monsoon seasons, potential traffic congestion in urban areas, road quality variances in rural sections, and the transport of hazardous materials. Mitigation strategies should focus on timely weather monitoring, traffic prediction, adherence to safety regulations, and route planning to adjust for emergencies and less predictable rural road conditions. Regular communication checks and driver awareness programs will enhance safety and reduce risks associated with heavy vehicle transport on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Roundabout	High	11.63247, 78.12734	15 KM/Hr
2	U-Turn	High	11.6193714, 78.1124012	10 KM/Hr
3	U-Turn	High	11.618013, 78.1109532	10 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	High	11.44900, 77.87406	15 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45781, 77.85203	30 KM/Hr
9	Turn	High	11.46307, 77.84941	15 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45605, 77.81683	15 KM/Hr
12	Turn	Medium	11.53573, 77.95794	30 KM/Hr
13	Turn	Medium	11.62100, 78.10723	30 KM/Hr
14	Blind Spot	Blind Spot	11.62130, 78.10776	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
16	Turn	Medium	11.61235, 78.10495	30 KM/Hr
17	Blind Spot	Blind Spot	11.61923, 78.11225	10 KM/Hr
18	Blind Spot	Blind Spot	11.61817, 78.11102	10 KM/Hr
19	Blind Spot	Blind Spot	11.63306, 78.12731	10 KM/Hr
20	Turn	Medium	11.62873, 78.16541	30 KM/Hr
21	Turn	Medium	11.66095, 78.20881	30 KM/Hr
22	Turn	Medium	11.66604, 78.32509	30 KM/Hr
23	Turn	Medium	11.65154, 78.35007	30 KM/Hr
24	Turn	High	11.65458, 78.37988	15 KM/Hr
25	Turn	Medium	11.65492, 78.38012	30 KM/Hr
26	Turn	High	11.65745, 78.40584	15 KM/Hr
27	Turn	High	11.70451, 78.41563	15 KM/Hr
28	Turn	High	11.70569, 78.41336	15 KM/Hr
29	Turn	High	11.70650, 78.41352	15 KM/Hr
30	Turn	Medium	11.70685, 78.41417	30 KM/Hr
31	Turn	Medium	11.72308, 78.44878	30 KM/Hr
32	Turn	Medium	11.72329, 78.44994	30 KM/Hr
33	Turn	Medium	11.74456, 78.46609	30 KM/Hr
34	Turn	High	11.74541, 78.48961	15 KM/Hr
35	Turn	Medium	11.75356, 78.49168	30 KM/Hr
36	Turn	Medium	11.77176, 78.50554	30 KM/Hr
37	Turn	High	11.77985, 78.51619	15 KM/Hr
38	Blind Spot	Blind Spot	11.78349, 78.56977	10 KM/Hr
39	Turn	Medium	11.78167, 78.56890	30 KM/Hr
40	Turn	Medium	11.77927, 78.56903	30 KM/Hr
41	Turn	High	11.77850, 78.56974	15 KM/Hr
42	Turn	High	11.77910, 78.57191	15 KM/Hr
43	Turn	Medium	11.78761, 78.57441	30 KM/Hr
44	Turn	High	11.78789, 78.57500	15 KM/Hr
45	Turn	Medium	11.78808, 78.57979	30 KM/Hr
46	Turn	Medium	11.78844, 78.58003	30 KM/Hr
47	Turn	Medium	11.78861, 78.58069	30 KM/Hr
48	Turn	High	11.79112, 78.58335	15 KM/Hr
49	Turn	Medium	11.79125, 78.58665	30 KM/Hr
50	Blind Spot	Blind Spot	11.79265, 78.58820	10 KM/Hr
51	Blind Spot	Blind Spot	11.79217, 78.58824	10 KM/Hr
52	Turn	Medium	11.80538, 78.61003	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
53	Turn	High	11.80530, 78.61515	15 KM/Hr
54	Turn	High	11.81546, 78.62225	15 KM/Hr
55	Turn	High	11.81457, 78.62473	15 KM/Hr
56	Turn	Medium	11.79002, 78.63059	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
19	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
20	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
21	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
22	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
31	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium
32	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
33	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium
34	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium
35	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
36	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
37	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
38	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium
39	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
23	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
25	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
26	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
27	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
29	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63247, 78.12734



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6193714, 78.1124012



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.618013, 78.1109532



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62100, 78.10723



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61235, 78.10495



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61923, 78.11225



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61817, 78.11102



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63306, 78.12731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62873, 78.16541



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66095, 78.20881



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66604, 78.32509



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65154, 78.35007



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65458, 78.37988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65492, 78.38012



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65745, 78.40584



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70569, 78.41336



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70650, 78.41352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70685, 78.41417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72308, 78.44878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72329, 78.44994



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74456, 78.46609



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74541, 78.48961



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75356, 78.49168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77176, 78.50554



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77985, 78.51619



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.78349, 78.56977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78167, 78.56890



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77927, 78.56903



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77850, 78.56974



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77910, 78.57191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78761, 78.57441



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.78789, 78.57500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78808, 78.57979



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78844, 78.58003



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78861, 78.58069



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.79112, 78.58335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79125, 78.58665



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.79265, 78.58820



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.79217, 78.58824



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80538, 78.61003



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80530, 78.61515



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.81546, 78.62225
